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Studierichting: Informatica
Oefeningenreeks 7B nr. 1.3

Opgave: zoek $\frac{dw}{dt}$: $w = \sin xyz$ met $x = t$, $y = t^2$ en $z = t^3$

$\frac{\partial w}{\partial x} = \cos(xyz) \cdot \frac{\partial(xyz)}{\partial x} = \cos(xyz) \cdot (yz)$

$\frac{\partial w}{\partial y} = \cos(xyz) \cdot (xz)$

$\frac{\partial w}{\partial z} = \cos(xyz) \cdot (xy)$

$\frac{dx}{dt} = 1$

$\frac{dy}{dt} = 2t$

$\frac{dz}{dt} = 3t^2$

$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$

$= \cos(xyz) \cdot \left(yz + 2t \cdot xz + 3t^2 \cdot xy\right)$

$= \cos\left(t^6\right) \cdot \left(t^5 + 2t^5 + 3t^5\right) = \cos\left(t^6\right) \cdot \left(6t^5\right)$

$$\frac{dw}{dt} = \cos\left(t^6\right) \cdot \left(6t^5\right)$$

References